TARESH BANSAL

+91 9350426503 ◇ tareshbansal06@gmail.com

RESEARCH INTERESTS

IoT Security · Efficient Multimodal Models · Vision-Language Models · On-Device Deep Learning · LLM Agents

EDUCATION

Birla Institute of Technology and Science (BITS), Pilani, India Aug. 2025 – May 2027 (expected)
Master of Engineering in Computer Science **GPA: 8.68 / 10.0**
Thesis (in progress): "Token Pruning for VLMs" Supervisor: Prof. Vinay Chamola

Birla Institute of Technology and Science (BITS), Pilani, India Nov. 2020 – May 2025
Bachelor of Engineering in Computer Science **GPA: 8.13 / 10.0**
Master of Science in Chemistry (dual-degree program)

PUBLICATIONS

A Novel Competing Convolutional Kernels Method to CSI-Based Fall Detection for Disabled People [DOI]
Siva Sai, **Taresh Bansal**, Vinay Chamola
IEEE Internet of Things Magazine **(Q1, Impact Factor 5.8)** Nov. 2025

Drift-Resilient Intrusion Detection in IoMT Networks Using Adaptive Hoeffding Trees with Internal Node Reassessment
Taresh Bansal, Siva Sai, Vinay Chamola, Mohan Gurusamy
IEEE Internet of Things Magazine **(Q1, Impact Factor 5.8)**, under review Sep. 2026

RESEARCH EXPERIENCE

Visual Token Pruning for Medical Vision-Language Models Aug. 2026 – Present
BITS Pilani, Pilani, India

- Asked whether medical AI models really need all the image data they read, which drives their cost.
- Reproduced CDPruner (NeurIPS 2025) on two open models, matching all **24** scores within **0.32 points**.
- On natural-image benchmarks with LLaVA-1.5, accuracy peaked at just **32 of 576** image tokens.
- Ported it to LLaVA-Med v1.5 and ran a blank-image control over **2,512** medical questions.
- Found only one of the three medical benchmarks, SLAKE, clearly needed the image to answer.
- On SLAKE, CDPruner beat random selection by **3.9 points** (closed questions, $p = 0.004$).

Lightweight Architectures for Plant Disease Recognition on Edge Devices Mar. – Aug. 2026
BITS Pilani, Pilani, India

- Asked which small models are actually worth deploying on phones for crop disease detection.
- Compared **9** models under **2.2M** parameters across **3** datasets (**118k** images, up to 115 diseases).
- Found the proposed model (**2.03M** parameters) best overall, with top accuracy on **2 of 3** datasets.
- Led the next model by **3.8 points** on the hardest, 115-class dataset (**69.8%** against **66.0%**).
- Showed CBAM attention hurt rather than helped, costing **3.4 points** of accuracy on the hardest dataset.
- Measured **10.5 ms** per image over **200** runs, **40%** faster than the other Ghost-based models (17–18 ms).

Drift-Resilient Intrusion Detection for IoMT Networks Jan. – Feb. 2026

- Tackled attack detection on medical IoT devices, where traffic shifts over time and models go stale.
- Trained on the WUSTL-EHMS-2020 dataset, balancing its attack classes with **SMOTE** after cleaning.
- Reached **99.94%** accuracy, **9.05 points** above the best incremental baseline, at **100%** precision.
- Updated in **0.65 s** per cycle against **41.5 s** to retrain NuSVC and **199 s** for Isolation Forest.

Flood Detection from Drone Imagery Nov. – Dec. 2025

- Trained ResNet, InceptionNet and Xception to spot flooding in drone photos, reusing pretrained layers.
- Tuned the training settings (BCEWithLogitsLoss, Adam) to reach **99.1%** accuracy.

Image Inpainting with GANs and Autoencoders Feb. – Apr. 2025

- Filled missing image regions with MI-GAN and a custom autoencoder: **92.3%** and **84.9%** accuracy.

TEACHING EXPERIENCE

Teaching Assistant, BITS Pilani Aug. 2024 – Present

Introduction to Logic, BITS Pilani Digital	Sep. 2026 – Present
Introduction to LLMs, BITS Pilani	Aug. 2026 – Present
Computing Systems, BITS Pilani Digital	May 2026 – Aug. 2026
Introduction to Computing Systems, BITS Pilani Digital	Dec. 2025 – Mar. 2026
Object-Oriented Programming, BITS Pilani Digital	Sep. 2025 – Dec. 2025
Foundations of Data Science, BITS Pilani	Aug. 2025 – Dec. 2025
Machine Learning, BITS Pilani	Aug. 2024 – Dec. 2024

POSITIONS OF RESPONSIBILITY

Executive Committee Member, National Service Scheme (NSS) Aug. 2021 – July 2022
BITS Pilani, Pilani, India

- Managed village health and eye camps with a team of 20+ volunteers, serving 250+ attendees.
- Led a mental-health and technology-awareness lecture series for 180+ students with 30+ volunteers.
- Started the "Health OK Please" podcast on cervical cancer and emergency care, reaching 100+ people.

Event Coordinator, Chemistry Association Aug. 2021 – Apr. 2022
BITS Pilani, Pilani, India

- Ran **four** live events at APOGEE, the annual tech-fest: **1500+** registrations, **200%** more footfall.
- Tripled inter-college reach and collaborations by spacing events evenly, leading **40+** volunteers.

Anti-COVID Force, BITS Pilani Apr. 2021 – Aug. 2021

- Coordinated 100+ volunteers who resolved 400+ COVID-19 queries on oxygen, beds and medical facilities.

PROFESSIONAL EXPERIENCE

NLP Intern, Palmtree Infotech May 2022 – July 2022

- Built a language-conversion tool from English to Hebrew and other languages using Python and spaCy.
- Implemented NLP techniques to extract the key content of documents for use in multilingual settings.

TECHNICAL SKILLS

Languages	Python, C++, SQL
Machine Learning	PyTorch, Computer Vision
Data Processing	NumPy, Pandas, PySpark
LLM Tooling	LangChain, MCP, Pydantic

RELEVANT COURSEWORK

Machine Learning, Deep Learning, Data Mining, Probability and Statistics, Data Structures and Algorithms, Distributed Systems, Operating Systems, Database Systems, Object-Oriented Programming.

HONORS AND ACHIEVEMENTS

2026: Qualified the Graduate Aptitude Test in Engineering (GATE) in Computer Science.
2025: Received a full-time Software Development Engineer offer from Orange Business.